12.10) 반복이 없는 이원 배치 실험의 모형

$$y_{ij} = \mu + a_i + \beta_j + \epsilon_{ij} \qquad i = 1, 2, \cdots, l$$
$$j = 1, 2, \cdots, m$$

에서 ϵ_{ij}는 서로 독립이고 $N(0, \sigma^2)$의 분포를 따른다고 가정하자.

(1) $\bar{\epsilon}_{i\cdot} = \sum_{j=1}^{m} \epsilon_{ij}/m$, $\bar{\epsilon}_{\cdot j} = \sum_{i=1}^{l} \epsilon_{ij}/l$, $\bar{\bar{\epsilon}} = \sum_{i=1}^{l}\sum_{j=1}^{m} \epsilon_{ij}/lm$ 을 나타낼 때

$$\sum_{i=1}^{l}\sum_{j=1}^{m}(\bar{\epsilon}_{i\cdot} - \bar{\bar{\epsilon}})^2 \text{ 과 } \sum_{i=1}^{l}\sum_{j=1}^{m}(\epsilon_{ij} - \bar{\epsilon}_{i\cdot} - \bar{\epsilon}_{\cdot j} + \bar{\bar{\epsilon}})^2$$

이 서로 독립임을 증명하라.

(2) 만약 $l = 4$일 때 귀무가설

$$H_0 : a_1 = 2a_2 = 3a_3$$

을 검정하는 F-통계량을 구하여라.

(Solution)

(1) 을 풀수 있는 방법은 총 2가지이다: covariance=0 또는 코크란의 정리 (Cochran's Theorem)를 사용하는 방법이다.

$$U_i = \bar{\epsilon}_i - \bar{\bar{\epsilon}}$$
$$V_{kj} = \epsilon_{kj} - \bar{\epsilon}_{k\cdot} - \bar{\epsilon}_{\cdot j} + \bar{\bar{\epsilon}}$$

WTS: $\mathrm{cov}(U_i, V_{kj}) = 0$

$$\mathrm{cov}(U_i, V_{kj}) = \mathrm{cov}(\bar{\epsilon}_{i\cdot} - \bar{\bar{\epsilon}}, \epsilon_{kj} - \bar{\epsilon}_{k\cdot} - \bar{\epsilon}_{\cdot j} + \bar{\bar{\epsilon}})$$
$$= \underbrace{\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \epsilon_{kj} - \bar{\epsilon}_{k\cdot} - \bar{\epsilon}_{\cdot j} + \bar{\bar{\epsilon}})}_{*} - \underbrace{\mathrm{cov}(\bar{\bar{\epsilon}}, \epsilon_{kj} - \bar{\epsilon}_{k\cdot} - \bar{\epsilon}_{\cdot j} + \bar{\bar{\epsilon}})}_{*}$$

공분산 계산 규칙:

- $\mathrm{cov}(\epsilon_{ij}, \epsilon_{ij}) = \sigma^2$
- $\mathrm{cov}(\bar{\epsilon}_i, \epsilon_{ij}) = \frac{1}{m}\sum_{p=1}^{m} \mathrm{cov}(\epsilon_{ip}, \epsilon_{ij}) = \frac{1}{m}\mathrm{cov}(\epsilon_{ij}, \epsilon_{ij}) = \frac{\sigma^2}{m}$
- $\mathrm{cov}(\bar{\epsilon}_{\cdot j}, \epsilon_{ij}) = \frac{1}{l}\sum_{p=1}^{l} \mathrm{cov}(\epsilon_{ip}, \epsilon_{ij}) = \frac{1}{\ell}\mathrm{cov}(\epsilon_{ij}, \epsilon_{ij}) = \frac{\sigma^2}{\ell}$
- $\mathrm{cov}(\bar{\bar{\epsilon}}, \epsilon_{ij}) = \mathrm{cov}(\bar{\bar{\epsilon}}, \bar{\epsilon}_{i\cdot}) = \mathrm{cov}(\bar{\bar{\epsilon}}, \bar{\epsilon}_{\cdot j}) = \mathrm{cov}(\bar{\bar{\epsilon}}, \bar{\bar{\epsilon}}) = \frac{\sigma^2}{m\ell}$

$\}$ *

- $\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \epsilon_{ij}) = \sigma^2/m$
- $\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \bar{\epsilon}_{i\cdot}) = \sigma^2/m$
- $\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \bar{\epsilon}_{\cdot j}) = \sigma^2/ml$
- $\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \bar{\bar{\epsilon}}) = \sigma^2/ml$

$\}$ * (case 1: $i = k$ 인 경우) 같은 행

- $\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \epsilon_{kj}) = 0$
- $\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \epsilon_{k\cdot}) = 0$
- $\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \bar{\epsilon}_{\cdot j}) = \sigma^2/lm$
- $\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \bar{\bar{\epsilon}}) = \sigma^2/lm$

$\}$ * (case 2: $i \neq k$ 인 경우) 다른 행

$$\therefore \mathrm{cov}(\bar{\epsilon}_{i\cdot}, \epsilon_{kj} - \bar{\epsilon}_{k\cdot} - \bar{\epsilon}_{\cdot j} + \bar{\bar{\epsilon}}) = \sigma^2 m^{-1} - \sigma^2 m^{-1} - \sigma^2 m^{-1} l^{-1} + \sigma^2 m^{-1} \ell^{-1} = 0 \quad (\text{if } i = k)$$
$$\mathrm{cov}(\bar{\epsilon}_{i\cdot}, \epsilon_{kj} - \bar{\epsilon}_{k\cdot} - \bar{\epsilon}_{\cdot j} + \bar{\bar{\epsilon}}) = 0 - 0 - \sigma^2/ml + \sigma^2/ml = 0 \quad (\text{if } i \neq k)$$
$$\mathrm{cov}(\bar{\bar{\epsilon}}, \epsilon_{kj} - \bar{\epsilon}_{k\cdot} - \bar{\epsilon}_{\cdot j} + \bar{\bar{\epsilon}}) = \sigma^2 m^{-1} l^{-1} - \sigma^2 m^{-1} l^{-1} - \sigma^2 m^{-1} l^{-1} + \sigma^2 m^{-1} l^{-1} = 0$$

$$\therefore \mathrm{cov}(U_i, V_{kj}) = 0 \quad \forall\, i, j, k \in (1, \cdots, n)$$

$\Rightarrow$ Arbitrary functions $f(U_i)$ and $g(V_{kj})$ are also independent.

Cochran's Theorem Approach:

$$\epsilon \sim N(0, \sigma^2 I_{lm})$$

표평균을 나타내는 사영행렬 (Projection Matrix):

J_K: 모든 원소가 1인 $K \times K$ 행렬
$\otimes$: 크로네커 곱 (Kronecker product)

총평균 사영행렬 (P_0): $\mathbb{1}_{lm}(\mathbb{1}_{lm}^T \mathbb{1}_{lm})^{-1}\mathbb{1}_{lm}^T = \frac{1}{lm} J_{lm}$

행평균 사영행렬 (P_A): $\frac{1}{m}(I_l \otimes J_m)$

열평균 사영행렬 (P_B): $\frac{1}{l}(J_l \otimes I_m)$

2. 제곱합의 이차형식 표현

$$Q_1 = \sum_{i=1}^{l}\sum_{j=1}^{m}(\bar{\epsilon}_{i\cdot} - \bar{\bar{\epsilon}})^2 = \epsilon'(P_A - P_0)\epsilon = \epsilon' A_1 \epsilon$$
, 여기서 $A_1 = P_A - P_0$ 입니다.

두 번째 식 (잔차제곱 합의 성분):

$$Q_2 = \sum_{i=1}^{l}\sum_{j=1}^{m}(\epsilon_{ij} - \bar{\epsilon}_{i\cdot} - \bar{\epsilon}_{\cdot j} + \bar{\bar{\epsilon}})^2 = \epsilon'(I_{lm} - P_A - P_B + P_0)\epsilon = \epsilon' A_2 \epsilon$$

여기서 A_2는 $I_{lm} - P_A - P_B + P_0$ 입니다.

3. 사영행렬 간의 곱셈 성질

이차형식의 독립성을 증명하기 위해서는 행렬 A_1과 A_2의 곱이 영행렬(0)이 되어야 합니다. 이를 위해 사영행렬들 간의 곱을 먼저 계산해보겠습니다.

- 멱등성 (idempotency): $P_A P_A = P_A$
- 부분공간 포함 관계에 의한 곱: P_A 공간은 P_0 공간을 포함하므로, 두 행렬을 곱하면 더 작은 공간인 P_0가 나옵니다.

$$P_A P_0 = P_0 P_A = P_0$$

- 행과 열 사영행렬의 곱: 크로네카 곱의 성질 $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$를 이용합니다.

$$P_A P_B = \left(\frac{1}{m}(I_\ell \otimes J_m)\right)\left(\frac{1}{\ell}(J_\ell \otimes I_m)\right) = \frac{1}{lm}(I_l J_l \otimes I_m J_m)$$
$$= \frac{1}{lm}(J_l \otimes J_m)$$

즉, $P_A P_B = P_B P_A = P_0$ 입니다.

4. 직교성 (Orthogonality):

$$\begin{aligned}
A_1 A_2 &= (P_A - P_0)(I_{lm} - P_A - P_B + P_0) \\
&= P_A(I_{lm} - P_A - P_B + P_0) - P_0(I_{lm} - P_A - P_B + P_0) \\
&= (P_A - P_A P_A - P_A P_B + P_0) - (P_0 - P_0 P_A - P_A P_B + P_A P_0) \\
&= (P_A - P_A - P_0 + P_0) - (P_0 - P_0 - P_0 + P_0) \\
&= 0 - 0 \\
&= 0
\end{aligned}$$

(2) 만약 $l=4$ 일 때 귀무가설

$$H_0 : a_1 = 2a_2 = 3a_3$$

을 검정하는 F-통계량을 구하여라.

$$(y_{ij})_{\substack{1\le i\le l \\ 1\le j\le m}} \in \mathbb{R}^{l\times m}, \quad (y_{ij}) = \mu J_{lm} + a \otimes \mathbb{1}_m^T + b^T \otimes \mathbb{1}_l$$

$$(y_{ij}) = \begin{bmatrix} \mathbb{1}_{lm} & I_l \otimes \mathbb{1}_m & I_m \otimes \mathbb{1}_J \end{bmatrix} \begin{bmatrix} \mu & a_1 & \cdots & a_l & \beta_1 & \cdots & \beta_m \end{bmatrix}^T$$

$l=4$, $m=2$ 라고 가정하자.

$$\begin{bmatrix} y_{11} \\ \vdots \\ y_{41} \\ y_{12} \\ \vdots \\ y_{42} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ \beta_1 \\ \beta_2 \end{bmatrix} \Rightarrow y = X\beta, \quad y \in \mathbb{R}^8, \quad X \in \mathbb{R}^{8\times 7}, \quad \beta = \mathbb{R}^{7\times 1}$$

(Reduced Model)

$$\begin{bmatrix} y_{11} \\ \vdots \\ y_{41} \\ y_{12} \\ \vdots \\ y_{42} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & \frac{1}{2} & 0 & 1 & 0 \\ 1 & \frac{1}{2} & 0 & 0 & 1 \\ 1 & 1/3 & 0 & 1 & 0 \\ 1 & 1/3 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ a_1 \\ a_4 \\ \beta_1 \\ \beta_2 \end{bmatrix} \Rightarrow y = X_r \beta_r, \quad X_r \in \mathbb{R}^{8\times 5}, \quad \beta_r \in \mathbb{R}^{5\times 1}$$

$$F = \frac{\{SSE(R) - SSE(F)\}/2}{SSE(F)/(n-p+1)} = \frac{\{y^T(I_n - \Pi_{X_r})y - y^T(I_n - \Pi_X)y\}/2}{y^T(I_n - \Pi_X)y/p}$$, $p+1$: number parameter coefficients.

SST/(n-p-1)?

SSE(R) : Sum of squared error/residual in reduced model

SSE(F) : Sum of squared error/residual in full model

$p = \text{rank}(X) = 1 + (4-1) + (m-1) = m+3$

↳ 1을 빼주는 이유 : a_1, a_2, a_3, a_4에 해당하는 4개의 열을 모두 더하면 모든 원소가 1인 벡터가 된다.

: $\beta_1, \cdots, \beta_m$에 해당하는 m개의 열을 모두 더해도 모든 원소가 1인 벡터가 된다.

∴ ~~$F \sim F(2, m+3)$~~

$$\begin{aligned}
n-p-1 &= l\cdot m - (1 + (l-1) + (m-1)) \\
&= 4m - l - m + 1 \\
&= 3m - 3 = 3(m-1)
\end{aligned}$$

$$X_\mu = \sum_{i=1}^{l} X_{a_i} = \sum_{j=1}^{m} X_{\beta_j}$$

12.11 반복이 있는 이원배치실험의 모형

$$y_{ijk} = \mu + a_i + \beta_j + (a\beta)_{ij} + \epsilon_{ijk}$$

$i = 1, 2, \cdots, l$

$j = 1, 2, \cdots, m$

$k = 1, 2, \cdots, r$

에서 ϵ_{ijk}는 서로 독립이고 $N(0, \sigma^2)$의 분포를 따른다고 한다.

(1) a_i 와 β_j의 최소제곱추정량은 서로 독립임을 증명하여라.

(2) 만약 $m=3$ 일 때에 귀무가설

$$H_0 : \beta_1 = \beta_2$$

을 F-검정하는 통계량을 구하여라.

p.399-400을 참조

$$\hat{\mu} = \bar{\bar{y}} = \frac{1}{l\cdot m\cdot r} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{r} y_{ijk}$$

$$\left.\begin{aligned}
&\hat{t}_i = \bar{y}_{i..} - \bar{\bar{y}}, \quad \bar{y}_{i..} = \frac{1}{m\cdot r} \sum_{j=1}^{m} \sum_{k=1}^{r} y_{ijk} \\
&\hat{\beta}_j = \bar{y}_{.j.} - \bar{\bar{y}}, \quad \bar{y}_{.j.} = \frac{1}{l\cdot r} \sum_{i=1}^{l} \sum_{k=r}^{r} y_{ijk} \\
&\widehat{(t\beta)}_{ij} = \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{\bar{y}}
\end{aligned}\right\} \text{Memorize}$$

If $\hat{t}_i$ and $\hat{\beta}_j$ are independent, $\langle \hat{t}_i, \hat{\beta}_j \rangle = 0$

(1) a_i 와 β_j의 최소제곱추정량은 서로 독립임을 증명하여라.

$$\begin{aligned}
\text{cov}(\hat{a}_i, \hat{\beta}_j) &= \text{cov}(\bar{y}_{i..} - \bar{\bar{y}}, \bar{y}_{.j.} - \bar{\bar{y}}) \\
&= \underbrace{\text{cov}(\bar{y}_{i..}, \bar{y}_{.j.})}_{①} - \underbrace{\text{cov}(\bar{y}_{i..}, \bar{\bar{y}})}_{②} - \underbrace{\text{cov}(\bar{\bar{y}}, \bar{y}_{.j.})}_{③} + \underbrace{\text{cov}(\bar{\bar{y}}, \bar{\bar{y}})}_{④}
\end{aligned}$$

① $= \text{cov}\left(\frac{1}{m\cdot r} \sum_j \sum_k y_{ijk}, \frac{1}{l\cdot r} \sum_i \sum_k y_{ijk}\right)$

$= \frac{1}{m\cdot l\cdot r^2} \text{cov}\left(\sum_k y_{ijk}, \sum_k y_{ijk}\right) = \frac{1}{m\cdot l\cdot r} \sigma^2$

* Caution : k', k''과 같은 기호를 사용하여 구분 지을 것.

② $= \text{cov}\left(\frac{1}{mr} \sum_j \sum_k y_{ijk}, \frac{1}{l\cdot m\cdot r} \sum_i \sum_j \sum_k y_{ijk}\right)$

$= \frac{1}{l\cdot m^2\cdot r^2} \cdot m\cdot r \cdot \text{cov}(y_{ijk}, y_{ijk}) = \frac{1}{l\cdot m\cdot r} \sigma^2$

③ $= \text{cov}\left(\frac{1}{l\cdot r} \sum_i \sum_k y_{ijk}, \frac{1}{l\cdot m\cdot r} \sum_i \sum_j \sum_k y_{ijk}\right) = \cdots = \frac{1}{l\cdot m\cdot r} \sigma^2$

④ $= \text{cov}\left(\frac{1}{l\cdot m\cdot r} \sum_i \sum_j \sum_k y_{ijk}, \frac{1}{l\cdot m\cdot r} \sum_i \sum_j \sum_k y_{ijk}\right)$

$= \left(\frac{1}{l\cdot m\cdot r}\right)^2 l\cdot m\cdot r \, \text{cov}(y_{ijk}, y_{ijk}) = \frac{1}{l\cdot m\cdot r} \sigma^2$

∴ ① - ② - ③ + ④ $= \frac{1}{l\cdot m\cdot r}\sigma^2 - \frac{1}{l\cdot m\cdot r}\sigma^2 - \frac{1}{l\cdot m\cdot r}\sigma^2 + \frac{1}{l\cdot m\cdot r}\sigma^2 = 0$

∴ $\text{cov}(\hat{a}_i, \hat{\beta}_j) = 0 \Leftrightarrow \hat{a}_i$ and $\hat{\beta}_j$ are independent. ✓✓

(2) 만약 $m=3$ 일 때에 귀무가설

$H_0 : \beta_1 = \beta_2$

을 F-검정하는 통계량을 구하여라.

$l = 2$ 라고 가정하자

$$\begin{array}{cccccccccccc} \mu & \alpha_1 & \alpha_2 & \beta_1 & \beta_2 & \beta_3 & (\alpha\beta)_{11} & (\alpha\beta)_{12} & (\alpha\beta)_{13} & (\alpha\beta)_{21} & (\alpha\beta)_{22} & (\alpha\beta)_{23} \end{array}$$

$$\begin{bmatrix} 1&1&0&1&0&0&1&0&0&0&0&0 \\ 1&1&0&0&1&0&0&1&0&0&0&0 \\ 1&1&0&0&0&1&0&0&1&0&0&0 \\ 1&0&1&1&0&0&0&0&0&1&0&0 \\ 1&0&1&0&1&0&0&0&0&0&1&0 \\ 1&0&1&0&0&1&0&0&0&0&0&1 \\ 1&1&0&1&0&0&1&0&0&0&0&0 \\ 1&1&0&0&1&0&0&1&0&0&0&0 \\ 1&1&0&0&0&1&0&0&1&0&0&0 \\ 1&0&1&1&0&0&0&0&0&1&0&0 \\ 1&0&1&0&1&0&0&0&0&0&1&0 \\ 1&0&1&0&0&1&0&0&0&0&0&1 \end{bmatrix} = X$$

교호작용 (interaction)이 포함된 모형의 설계 행렬 (Design matrix)에서는 $\text{rank}(x) = l \cdot m$ 이 된다.

$$\begin{array}{cccccccccccc} \mu & \alpha_1 & \alpha_2 & \beta_1 & \beta_2 & \beta_3 & (\alpha\beta)_{11} & (\alpha\beta)_{12} & (\alpha\beta)_{13} & (\alpha\beta)_{21} & (\alpha\beta)_{22} & (\alpha\beta)_{23} \end{array}$$

$$\begin{bmatrix} 1&1&0&1&0&0&1&0&0&0 \\ 1&1&0&0&1&0&1&0&0&0 \\ 1&1&0&0&0&1&0&1&0&0 \\ 1&0&1&1&0&0&0&0&1&0 \\ 1&0&1&0&1&0&0&0&1&0 \\ 1&0&1&0&0&1&0&0&0&1 \\ 1&1&0&1&0&0&1&0&0&0 \\ 1&1&0&0&1&0&1&0&0&0 \\ 1&1&0&0&0&1&0&1&0&0 \\ 1&0&1&1&0&0&0&0&1&0 \\ 1&0&1&0&1&0&0&0&1&0 \\ 1&0&1&0&0&1&0&0&0&1 \end{bmatrix} = X_r$$

$$F = \frac{\{SSE(R) - SSE(F)\}/q}{SSE(F)/(n - \text{rank}(x))}$$

$$= \frac{y^T(I_n - \Pi_{X_r})y - y^T(I_n - \Pi_X)y / 1}{y^T(I_n - \Pi_X)y / \{lmr - ml\}}$$

$$= \frac{y^T(\Pi_X - \Pi_{X_r})y}{y^T(I_n - \Pi_X)y / \{3lr - 3l\}} \sim F(1, 3l(r-1))$$

$\Pi_X : X(X^TX)^{-1}X^{-1}$

y : response variable

I_n : Identity matrix

SSE : sum of squared residuals / error

R : reduced model

F : Full model

row-mean projection : $\frac{1}{c}(I_r \otimes J_c)$

$$\left[\begin{array}{ccc|ccc|ccc} 1&1&1&0&0&0&0&0&0 \\ 1&1&1&0&0&0&0&0&0 \\ 1&1&1&0&0&0&0&0&0 \\ \hline 0&0&0&1&1&1&0&0&0 \\ 0&0&0&1&1&1&0&0&0 \\ 0&0&0&1&1&1&0&0&0 \\ \hline 0&0&0&0&0&0&1&1&1 \\ 0&0&0&0&0&0&1&1&1 \\ 0&0&0&0&0&0&1&1&1 \end{array}\right] = I_3 \otimes J_3$$

Column mean projection : $\frac{1}{r}(J_r \otimes I_c)$

$$\left[\begin{array}{ccc|ccc|ccc} 1&0&0&1&0&0&1&0&0 \\ 0&1&0&0&1&0&0&1&0 \\ 0&0&1&0&0&1&0&0&1 \\ \hline 1&0&0&1&0&0&1&0&0 \\ 0&1&0&0&1&0&0&1&0 \\ 0&0&1&0&0&1&0&0&1 \\ \hline 1&0&0&1&0&0&1&0&0 \\ 0&1&0&0&1&0&0&1&0 \\ 0&0&1&0&0&1&0&0&1 \end{array}\right] = J_3 \otimes I_3$$

* 기억하기 용이하게 :

1. 순서 : $r \to c$
2. 행 (row) 평균 : I_r
   열 (column) 평균 : I_c
3. J_a : a로 나눔.
   예: $\frac{1}{c}(I_r \otimes J_c)$, $\frac{1}{r}(J_r \otimes I_c)$
   열 평균 　 행 평균.

12.11

$C_l = I_l - P_l$, $P_l = \frac{1}{l} J_l = \frac{1}{l} \mathbb{1}_l \mathbb{1}_l^T$

$\hat{\alpha}_i = \bar{y}_{i\cdot\cdot} - \bar{\bar{y}} = (C_l \otimes P_m \otimes P_r) y$

$\hat{\beta}_i = \bar{y}_{\cdot j \cdot} - \bar{\bar{y}} = (P_l \otimes C_m \otimes P_r) y$

$\widehat{(\alpha\beta)}_{ij} = \bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j \cdot} + \bar{\bar{y}} = (C_l \otimes C_m \otimes P_r) y$

$\hat{\epsilon} = (I_l \otimes I_m \otimes C_r) y$

$\hat{\alpha}_i$ and $\hat{\beta}_i$ are independent iff

$(C_l \otimes P_m \otimes P_r)(P_l \otimes C_m \otimes P_r) = 0$

$= (C_l P_l \otimes P_m C_m \otimes P_r P_r)$

$= (0 \otimes 0 \otimes P_r) = 0$

$\therefore$ $\hat{\alpha}_i$ and $\hat{\beta}_i$ are independent.